Class 11 Bioinformatics pt2

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The EBI Alphafold database

the EBI alphafold database contains lots of cmpured strucutres models. It is increaslilgy likely that the strcuture you are intrestee in is already in database at < https://alphafold.ebi.ac.uk >

there are three major outputs from Alphafold

1. A model of strcuture in **PDB** format.
2. A **pLDDT** score: that tells us how confident the model is for a given residue in your protein (High values are good, above70).
3. A **PAE** score taht tell us about protein packing quality

If you cant find a matching entry for the sequence you are intrested in AFDB you can run AlphaFold yourself…

Running AlphaFold

we will use ColabFold to run AlphaFold on our sequence < https://github.com/sokrypton/ColabFold >

Inerpreting Results

Custom analysis of resulting models

we can read all the AphaFold results into R and do more quanitative analysis than just viewing the structure in Mol-star:

The PDB database (the main respositriy of experiment strcuture) only has ~250 thousand structures (we saw this in the last lab). The main sequence database has over **200 million** sequneces! only 0.125% of known sequences have a known a structure - this is called the "strcuture knowledge gap".

```r
(250000 / 200000000)*100
```

```
[1] 0.125
```

Structures are much harder to determine than sequences they are expensive on avregae (about $1 million dollars each) they take on avrega 2-5 years to solveThe EBI has a database of pre-computed AlphFold (AF) models called AFDB. This is growing over time and it is useful to check before running AF ourselves.

Read all the PBD models:

```r
library(bio3d)
p<- read.pdb("hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_
```

```r
results_dir <- "hivpr_23119/"
```

```r
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

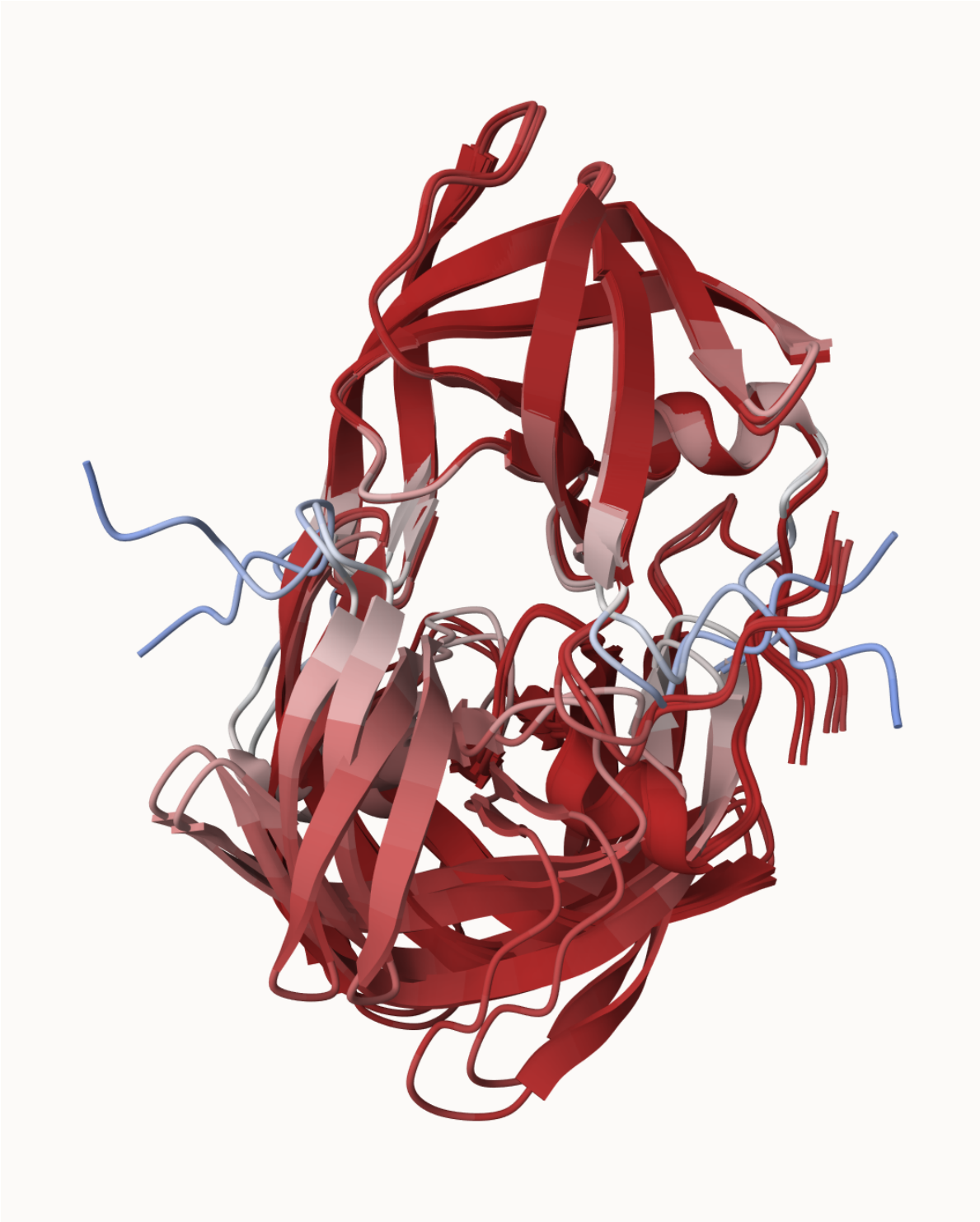
```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
```

```
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```r
knitr::include_graphics("HIVPR_23119_UNRELAXED_RANK_005_ALPHAFOLD2_MULTIMER_V3_MODEL_3_SEED_0
```



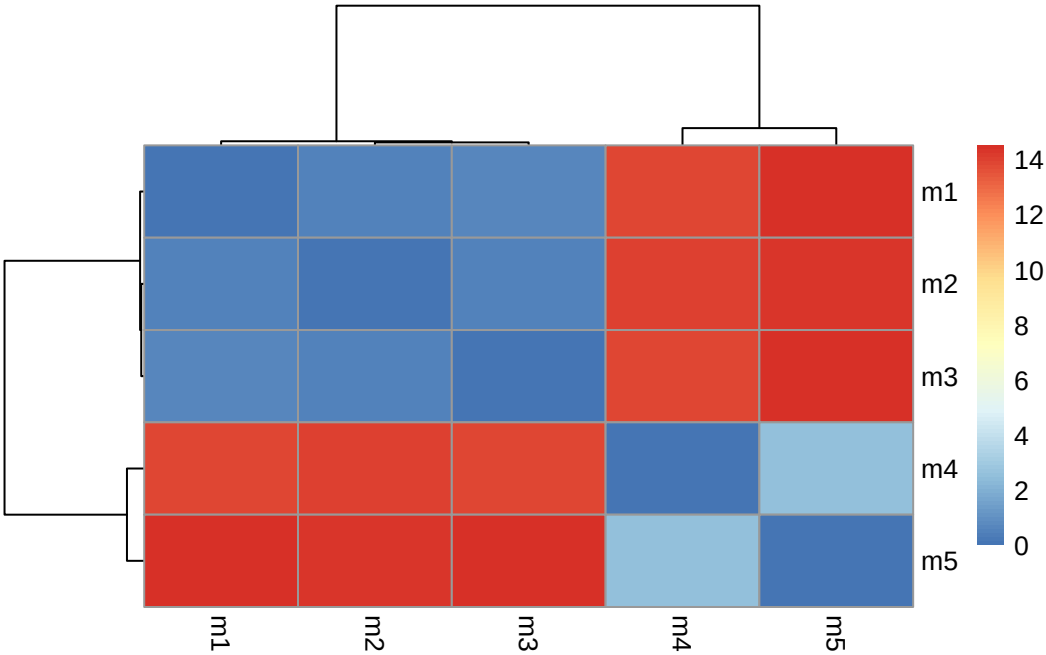
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.526
```

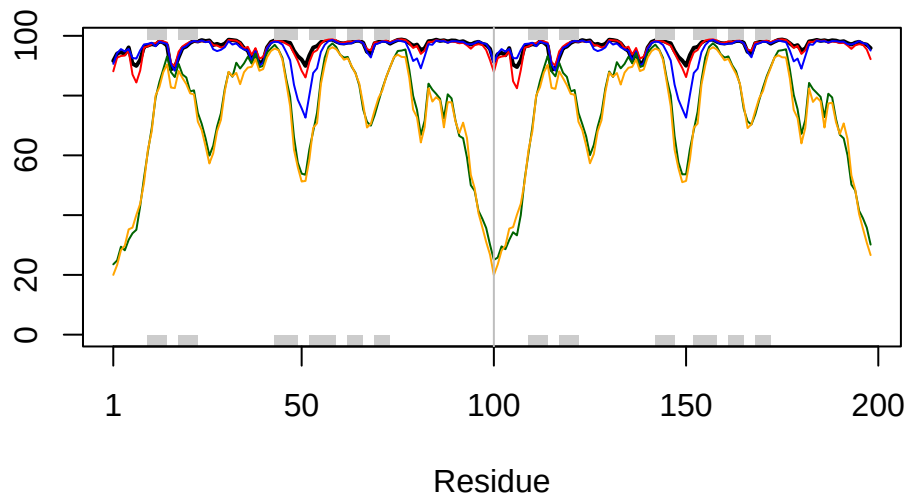
```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note:  Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 5437.294
core size 196 of 198  vol = 4705.336
core size 195 of 198  vol = 1827.704
core size 194 of 198  vol = 1121.539
core size 193 of 198  vol = 1047.76
core size 192 of 198  vol = 999.32
core size 191 of 198  vol = 953.718
core size 190 of 198  vol = 910.755
core size 189 of 198  vol = 870.203
core size 188 of 198  vol = 836.304
core size 187 of 198  vol = 805.237
core size 186 of 198  vol = 775.99
core size 185 of 198  vol = 752.564
```

```
core size 184 of 198  vol = 712.023
core size 183 of 198  vol = 685.568
core size 182 of 198  vol = 663.911
core size 181 of 198  vol = 645.881
core size 180 of 198  vol = 627.97
core size 179 of 198  vol = 611.812
core size 178 of 198  vol = 595.931
core size 177 of 198  vol = 581.132
core size 176 of 198  vol = 566.736
core size 175 of 198  vol = 548.587
core size 174 of 198  vol = 534.114
core size 173 of 198  vol = 505.214
core size 172 of 198  vol = 491.225
core size 171 of 198  vol = 473.905
core size 170 of 198  vol = 460.426
core size 169 of 198  vol = 444.81
core size 168 of 198  vol = 431.661
core size 167 of 198  vol = 421.542
core size 166 of 198  vol = 405.601
core size 165 of 198  vol = 392.666
core size 164 of 198  vol = 381.077
core size 163 of 198  vol = 367.559
core size 162 of 198  vol = 358.379
core size 161 of 198  vol = 346.865
core size 160 of 198  vol = 334.809
core size 159 of 198  vol = 324.09
core size 158 of 198  vol = 312.153
core size 157 of 198  vol = 301.296
core size 156 of 198  vol = 290.431
core size 155 of 198  vol = 281.319
core size 154 of 198  vol = 272.529
core size 153 of 198  vol = 263.215
core size 152 of 198  vol = 253.54
core size 151 of 198  vol = 240.86
core size 150 of 198  vol = 227.447
core size 149 of 198  vol = 215.581
core size 148 of 198  vol = 202.041
core size 147 of 198  vol = 195.426
core size 146 of 198  vol = 188.721
core size 145 of 198  vol = 181.778
core size 144 of 198  vol = 173.615
core size 143 of 198  vol = 165.946
core size 142 of 198  vol = 156.117
```

```
core size 141 of 198  vol = 149.814
core size 140 of 198  vol = 143.616
core size 139 of 198  vol = 135.81
core size 138 of 198  vol = 127.851
core size 137 of 198  vol = 122.596
core size 136 of 198  vol = 117.203
core size 135 of 198  vol = 109.848
core size 134 of 198  vol = 104.812
core size 133 of 198  vol = 98.776
core size 132 of 198  vol = 94.799
core size 131 of 198  vol = 90.494
core size 130 of 198  vol = 87.403
core size 129 of 198  vol = 83.558
core size 128 of 198  vol = 79.08
core size 127 of 198  vol = 75.056
core size 126 of 198  vol = 71.238
core size 125 of 198  vol = 67.735
core size 124 of 198  vol = 64.289
core size 123 of 198  vol = 61.381
core size 122 of 198  vol = 57.515
core size 121 of 198  vol = 53.254
core size 120 of 198  vol = 48.654
core size 119 of 198  vol = 45.832
core size 118 of 198  vol = 41.819
core size 117 of 198  vol = 38.71
core size 116 of 198  vol = 36.294
core size 115 of 198  vol = 33.386
core size 114 of 198  vol = 30.472
core size 113 of 198  vol = 27.786
core size 112 of 198  vol = 25.403
core size 111 of 198  vol = 22.827
core size 110 of 198  vol = 21.106
core size 109 of 198  vol = 19.327
core size 108 of 198  vol = 17.796
core size 107 of 198  vol = 16.235
core size 106 of 198  vol = 14.508
core size 105 of 198  vol = 12.969
core size 104 of 198  vol = 11.834
core size 103 of 198  vol = 11.185
core size 102 of 198  vol = 10.298
core size 101 of 198  vol = 8.898
core size 100 of 198  vol = 7.813
core size 99 of 198  vol = 6.074
```

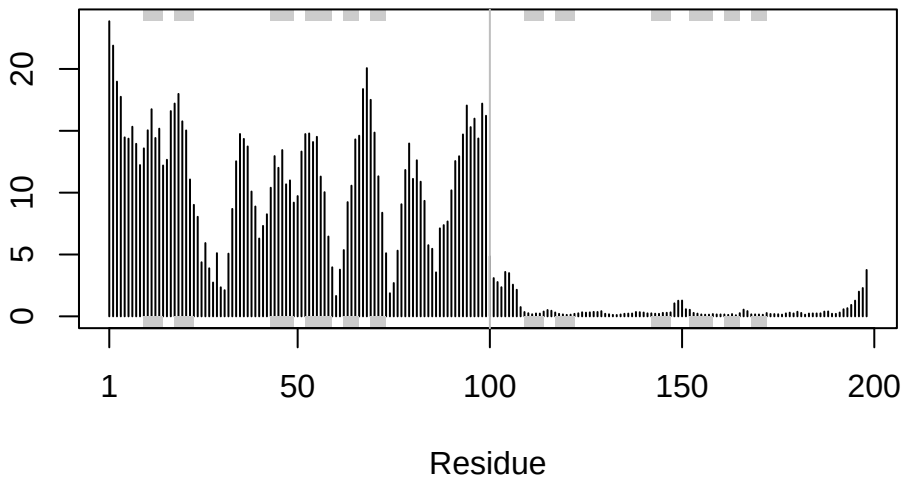
```
 core size 98 of 198  vol = 5.286
 core size 97 of 198  vol = 4.43
 core size 96 of 198  vol = 3.873
 core size 95 of 198  vol = 3.321
 core size 94 of 198  vol = 2.855
 core size 93 of 198  vol = 2.293
 core size 92 of 198  vol = 1.937
 core size 91 of 198  vol = 1.631
 core size 90 of 198  vol = 1.331
 core size 89 of 198  vol = 0.957
 core size 88 of 198  vol = 0.803
 core size 87 of 198  vol = 0.647
 core size 86 of 198  vol = 0.532
 core size 85 of 198  vol = 0.444
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 86 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     7   7      1
2     9  49     41
3    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```




```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.62 94.06 94.56 93.88 96.12 90.69
```

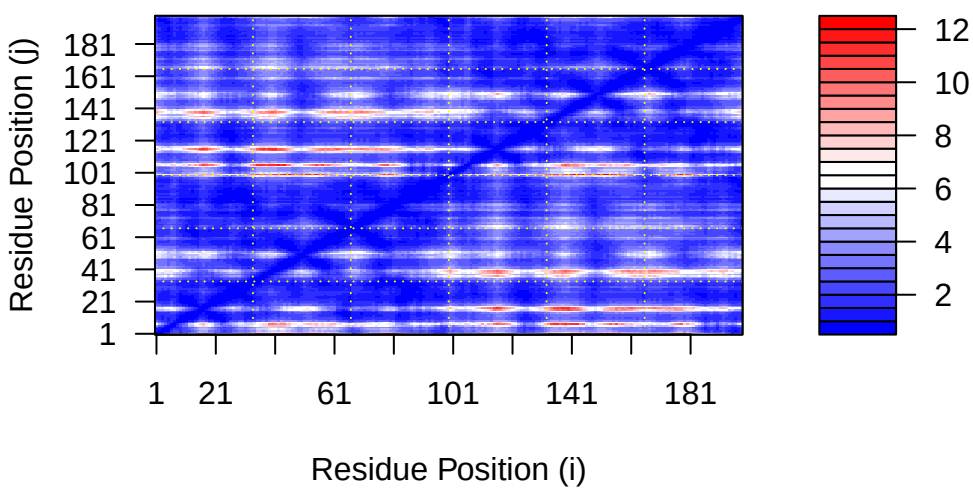
```r
pae1$max_pae
```

```
[1] 12.33594
```

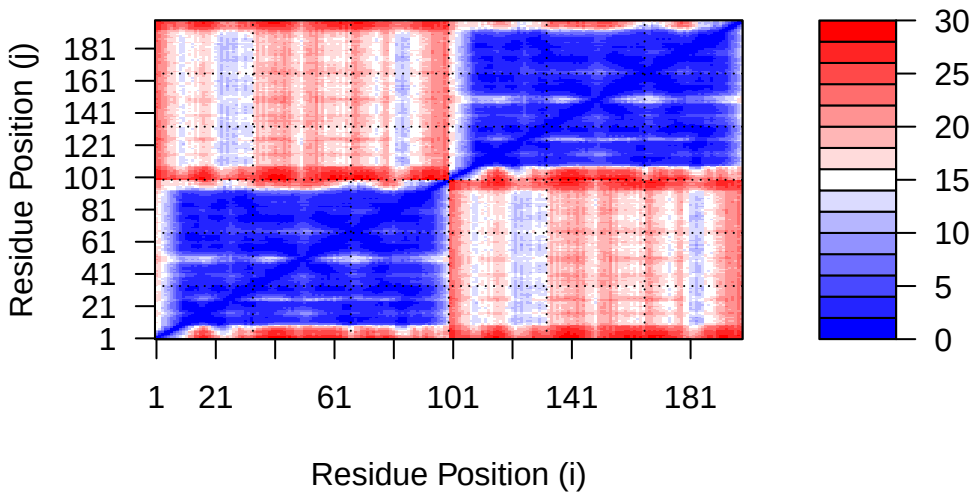
```r
pae5$max_pae
```

```
[1] 29.45312
```

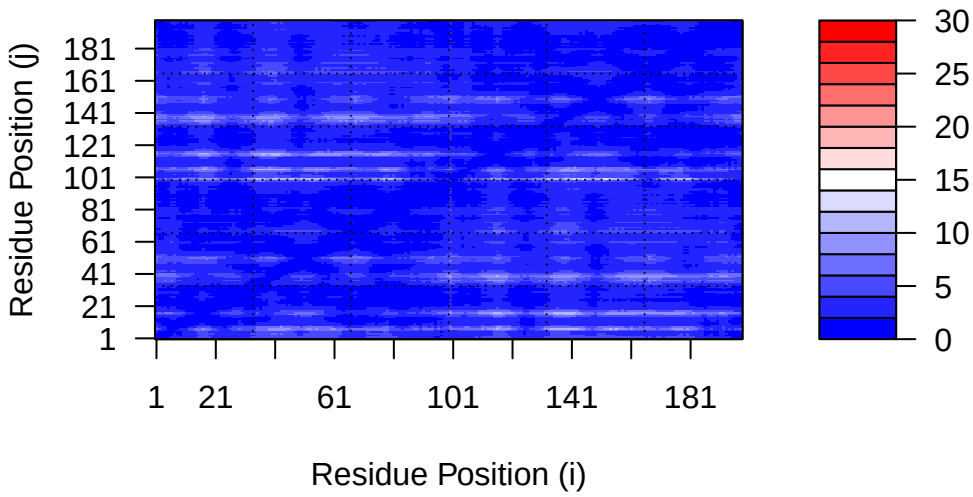
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

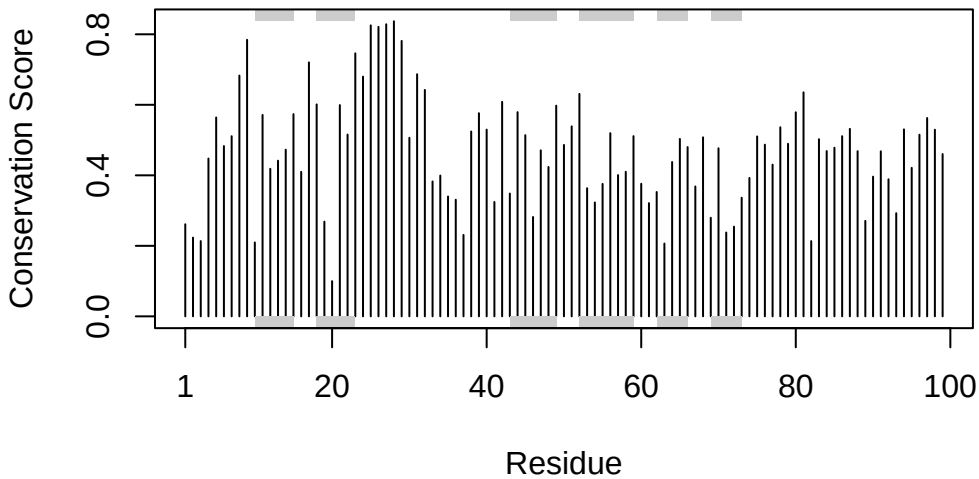
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

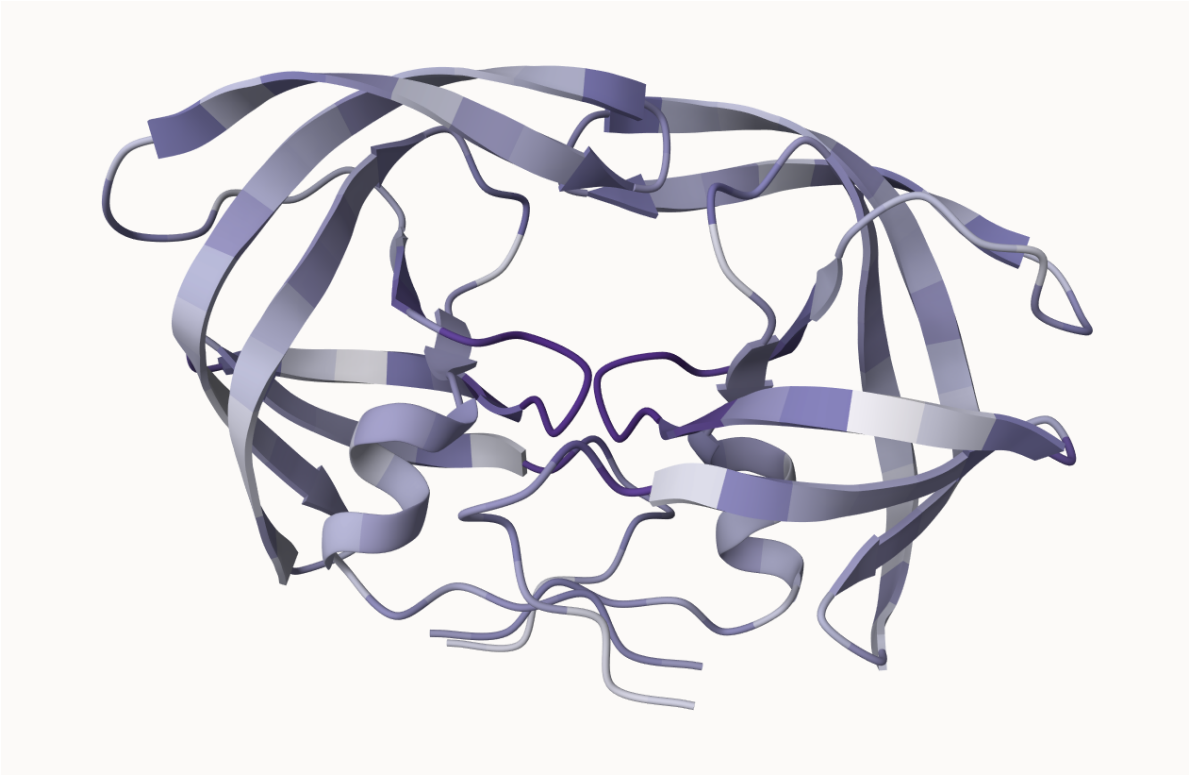
```
[1] 5397  132
```

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

```
knitr::include_graphics("M1_CONSERV.PDB.png")
```